Homework 4

1. Let X be a one-sided CGMY process for which $\log \mathbb{E}(e^{pX_t}) = t\int_0^\infty (e^{px} - 1 - px1_{x\le 1})\nu(x)dx$ for $p \in \mathbb{R}$, where $\nu(x) = \frac{Ce^{-Mx}}{x^{1+Y}}$ with $C, M > 0$ and $Y \in (1,2)$. Is $\mathbb{E}(e^{pX_t})$ finite for $p = M$? what about $p > M$?

Solution. Breaking up the integral above, we see that for $p = M$

$$\begin{aligned}
\int_1^\infty (e^{Mx} - 1)\frac{Ce^{-Mx}}{x^{1+Y}}dx &\le \int_1^\infty \frac{C}{x^{1+Y}}dx = \frac{C}{Y} < \infty \\
\int_0^1 (e^{Mx} - 1 - Mx)\frac{Ce^{-Mx}}{x^{1+Y}}dx &\le \int_0^1 \frac{1}{2}M^2e^Mx^2\frac{C}{x^{1+Y}}dx = \frac{const.}{2-Y} < \infty
\end{aligned}$$

where we have used that $(e^{Mx} - 1)e^{-Mx} \le 1$ in the first line, and $e^{Mx} - 1 - Mx \le \frac{1}{2}M^2e^Mx^2$ for $x \in [0,1]$ in the second line (which we can prove from the Taylor remainder theorem). Combining these two bounds, we see that $\mathbb{E}(e^{MX_t})$ is finite. For $p > M$, the first exponential in the first equation dominates the second exponential as $x \to \infty$, so $\mathbb{E}(e^{pX_t}) = \infty$. For the opposite reason, we also see that $\mathbb{E}(e^{pX_t}) < \infty$ for all $p < M$.

2. Let X be a Lévy process with Lévy density $\nu(x)$. What is the distribution of the number of jumps of size $\ge a$ over $[0,t]$? What is the expected length of time for the first positive jump of size $\ge a$ (assume $a > 0$).

Solution. From Lecture 2, the distribution is Poisson with parameter λt where $\lambda = \nu([a,\infty)) := \int_a^\infty \nu(x)dx$, and the times between such jumps are i.i.d. $\mathrm{Exp}(\lambda)$, so the expected time is $\frac{1}{\lambda}$.

3. Explain how can approximately simulate a Lévy process with positive-only jumps with a compound Poisson process.

Solution. Again from Lecture 2, we can use a compound Poisson process with rate $\lambda_\epsilon = \int_\epsilon^\infty \nu(x)dx$ and jump size density $\mu_\epsilon(x) = \frac{\nu(x)1_{x\ge\epsilon}}{\lambda_\epsilon}$. This is useful for simulating the CGMY process.

4. Let W denote a standard Brownian motion and $X_t = \gamma t + W_t$ for $\gamma \in \mathbb{R}$, and set $H_b := \min\{t : X_t = b\}$ for $b > 0$. Using that

$$\mathbb{E}(e^{-qH_b}) = e^{b(\gamma - \sqrt{\gamma^2 + 2q})} \qquad (q > 0) \tag{1}$$

and letting $q \searrow 0$, compute $\mathbb{P}(H_b < \infty)$ when $\gamma \ge 0$ and $\gamma < 0$. Compute the distribution of $\bar{X}_\infty$ (see q5 to see where (1) comes from; specifically the exponent $\Phi(q) = -\gamma + \sqrt{\gamma^2 + 2q}$ is the positive root of $V(p) = \gamma p + \frac{1}{2}p^2$).

Solution. As $q \to 0$, $e^{-qH_b} \to 1$ a.s. if $H_b < \infty$, or 0 if $H_b = \infty$. The family of random variables $(e^{-qH_b})_{b\ge 0}$ is clearly bounded by 1; hence (using the bounded convergence thm) we can take the limit inside the expectation as

$$\lim_{q\to 0}\mathbb{E}(e^{-qH_b}) = \mathbb{E}(\lim_{q\to 0} e^{-qH_b}) = \mathbb{E}(1_{H_b<\infty}) = \mathbb{P}(H_b < \infty) = e^{b(\gamma - \sqrt{\gamma^2})}.$$

The right hand side simplifies to 1 when $\gamma \ge 0$, and $e^{-2b|\gamma|}$ if $\gamma < 0$, so $H_b < \infty$ a.s. when $\gamma \ge 0$ but $\mathbb{P}(H_b < \infty) = e^{-2b|\gamma|} < 1$ when $\gamma < 0$, since in this case the overall trend of X is negative i.e. in the opposite direction to the upper barrier at $b > 0$. Since $\{H_b < \infty\} = \{\bar{X}_\infty \ge b\}$, we see that

$$\mathbb{P}(\bar{X}_\infty \ge b) = e^{-2b|\gamma|}$$

so $\bar{X}_\infty \sim \mathrm{Exp}(2|\gamma|)$ when $\gamma < 0$.

5. Compute $\mathbb{E}(e^{iu\underline{X}_{e_q}})$ for $u \in \mathbb{R}$ for a Lévy process X with negative-only jumps and a non-zero Brownian component, where $e_q \sim \mathrm{Exp}(q)$ is independent of X. Simplify your formula for the case when X is Brownian motion.

Solution. $\mathbb{E}(e^{pX_t}) = e^{t(\gamma p + \frac{1}{2}\sigma^2p^2 + \int_{-\infty}^0 (e^{px} - 1 - px1_{|x|\le 1})\nu(x)dx)} = e^{V(p)t}$ for some (γ, σ, ν) where $V(p) = \gamma p + \frac{1}{2}\sigma^2p^2 + \int_{-\infty}^0 (e^{px} - 1 - px1_{|x|\le 1})\nu(x)dx$, and $V(p)$ is finite for $p \in [0,\infty)$ since $e^{px} - 1 - px1_{|x|\le 1}$ is bounded for $x \le 0$ if $p \ge 0$ and $O(x^2)$ as $x \to 0$. Then $\mathbb{E}(e^{pX_t - V(p)t}) = 1$, and (from the Optional Stopping theorem for bounded stopping times), we can replace t with the bounded (stopping) time $t \wedge H_b$ to get

$$\mathbb{E}(e^{pX_{t\wedge H_b} - V(p)(t\wedge H_b)}) = 1$$

and setting $V(p) = q$ with $q > 0$ (we can do this because V is continuous and $V''(p) > 0$ and $V(p) \to \infty$ as $p \to \infty$ because $\sigma > 0$), we see that

$$\mathbb{E}(e^{\Phi(q)X_{t\wedge H_b} - q(t\wedge H_b)}) = 1$$

where we take $\Phi(q)$ to be the positive root of $V(p) = q$. $e^{\Phi(q)X_{t\wedge H_b} - q(t\wedge H_b)} \leq e^{\Phi(q)b}$ for all t, so letting $t \to \infty$ and applying the bounded convergence theorem, we obtain

$$\mathbb{E}(e^{-qH_b}) \quad = \quad e^{-\Phi(q)b} \tag{2}$$

since $X_{H_b} = b$ as X has no positive jumps (which we have seen before for the case when X is Brownian motion where $V(p) = \frac{1}{2}p^2$ and hence $\Phi(q) = \sqrt{2q}$). Then

$$\mathbb{P}(\bar{X}_{e_q} > b) \quad = \quad \mathbb{P}(e_q > H_b) \quad = \quad \mathbb{E}(1_{e_q > H_b}) \quad = \quad \mathbb{E}(\mathbb{E}(1_{e_q > H_b}|H_b)) \quad = \quad \mathbb{E}(e^{-qH_b})$$

so $\bar{X}_{e_q} \sim \text{Exp}(\Phi(q))$. Hence

$$\mathbb{E}(e^{iu\bar{X}_{e_q}}) \quad = \quad \frac{\Phi(q)}{\Phi(q) - iu}$$

and we know that $\mathbb{E}(e^{iuX_{e_q}}) = \frac{q}{q-\psi(u)}$ where $\mathbb{E}(e^{iuX_t}) = e^{t\psi(u)}$; hence re-arranging the Wiener-Hopf formula we see that

$$\mathbb{E}(e^{iu\underline{X}_{e_q}}) \quad = \quad \frac{\mathbb{E}(e^{iuX_{e_q}})}{\mathbb{E}(e^{iu\bar{X}_{e_q}})} \quad = \quad \frac{q(\Phi(q) - iu)}{(q - \psi(u))\Phi(q)}\,.$$

We still require a double IFT in u and q to get from this to the density of $\underline{X}_t$ (see short Mathematica notebook at `https://martinforde.github.io/DensityofMinimumofBMdoubleIFT.nb`), but we only require a single IFT to get the density or cdf of H_b using (2) with $q = -iu$ (use the Gil-Pelaez formula to compute the cdf), and recall that $\mathbb{P}(H_b \leq t) = \mathbb{P}(\bar{X}_t \geq b)$.

If X is Brownian motion, $\psi(u) = -\frac{1}{2}u^2$ so $V(p) = \frac{1}{2}p^2$ so $\Phi(q) = \sqrt{2q}$, and setting $u = -ip$, we find that

$$\mathbb{E}(e^{p\underline{X}_{e_q}}) \quad = \quad f(q,p) \quad = \quad \frac{\sqrt{2q}}{\sqrt{2q} + p}$$

for $p > -\sqrt{2q}$, so $\int_0^\infty e^{-qt}\mathbb{E}(e^{p\underline{X}_t})dt = \frac{1}{q}f(q,p)$, because $\mathbb{E}(e^{p\underline{X}_{e_q}}) = \int_0^\infty qe^{-qt}\mathbb{E}(e^{p\underline{X}_t})dt$.

6. Compute $\Phi(q)$ in q5 for the case of a one-sided CGMY model with negative-only jumps for which $\mathbb{E}(e^{pX_t}) = e^{t((G+p)^Y - G^Y)}$ for $p > -G$.

Solution. $V(p) = (G+p)^Y - G^Y$ (solving $V(p) = q$ we find that

$$\Phi(q) \quad = \quad (q + G^Y)^{\frac{1}{Y}} - G \qquad (q \geq 0)\,.$$

7. Consider a **double exponential Kou** compound Poisson process with

$$\nu(x) \quad = \quad a\rho e^{-\rho x}1_{x>0} \; + \; \hat{a}\hat{\rho}e^{\hat{\rho}x}1_{x<0}$$

with $\rho, \hat{\rho} > 0$, i.e. where the jump sizes are drawn from a two-sided asymmetric exponential distribution. Then

$$\psi(z) \quad = \quad \int_{-\infty}^{\infty}(e^{zx} - 1)\nu(x)dx \quad = \quad \frac{az}{\rho - z} - \frac{\hat{a}z}{\hat{\rho} + z} \tag{3}$$

and

$$\begin{aligned}\mathbb{E}(e^{zX_{e_q}}) \quad = \quad \int_0^\infty qe^{-qt}e^{\psi(z)t}dt \quad = \quad \frac{q}{q - \psi(z)} \quad = \quad \frac{q}{q + \frac{az}{z-\rho} + \frac{\hat{a}z}{z+\hat{\rho}}} \quad &= \quad \frac{q(z-\rho)(z+\hat{\rho})}{q(z-\rho)(z+\hat{\rho}) + az(z+\hat{\rho}) + \hat{a}z(z-\rho)} \\ &= \quad \frac{q(z-\rho)(z+\hat{\rho})}{(a+\hat{a}+q)(z-\zeta)(z+\hat{\zeta})}\end{aligned}$$

where ζ and $-\hat{\zeta}$ are the **roots** of $q - \psi(z)$, and the $a + \hat{a} + q$ comes from collecting all coefficients of z^2. We can re-write this in the form

$$c_1\,\frac{1 - \frac{z}{\rho}}{1 - \frac{z}{\zeta}} \cdot \frac{1 + \frac{z}{\hat{\rho}}}{1 + \frac{z}{\hat{\zeta}}} \tag{4}$$

for some constant c_1, and it turns out that c_1 simplifies to 1 (proof not required), and (4) gives us the **Wiener-Hopf** decomposition $\mathbb{E}(e^{zX_{e_q}}) = \mathbb{E}(e^{z\bar{X}_{e_q}}) \cdot \mathbb{E}(e^{z\underline{X}_{e_q}})$ for X. Multiplying $\psi(z) = q$ by $(\rho - z)(z + \hat{\rho})$ and using (3) we get

$$\psi(z)(\rho - z)(z + \hat{\rho}) \quad = \quad az(z + \hat{\rho}) \; - \; \hat{a}z(\rho - z) \quad = \quad (\rho - z)(z + \hat{\rho})q$$

which is just a quadratic $Q(z)$ in z, so ζ and $-\hat{\zeta}$ are just the positive and negative roots of this quadratic, which in general also depend on q.

8. Consider a Lévy process with

$$\nu(x) \quad = \quad \frac{e^{-Mx}}{(1-e^{-x})^{1+Y}} 1_{x>0}$$

for $x > 0$. Show that ν behaves similarly to the CGMY process for $x \ll 1$ and $x \gg 1$. Characterize the WH decomposition for this process.

Solution. $\nu(x) \sim \frac{e^{-Mx}}{x^{1+Y}}$ as $x \to 0$ (same as CGMY) and $\nu(x) \sim e^{-Mx}$ as $x \to \infty$ (similar to CGMY but without the power function $\frac{1}{x^{1+Y}}$). Using the binomial theorem, we find that

$$\nu(x) \quad = \quad e^{-Mx}(1 + (1+Y)e^{-x} + \frac{1}{2!}(1+Y)(2+Y)e^{-2x} + \frac{1}{3!}(1+Y)(2+Y)(3+Y)e^{-3x} + ...)$$

for $x > 0$, i.e. an infinite series of Kou-type Lévy densities, and we can write out a similar series for $\frac{e^{Gx}}{(1-e^{x})^{1+Y}}$ for $x < 0$. Then extending the analysis from the previous question to the two infinite series here for ν, we find that the WH decomposition takes the form

$$\prod_{n=1}^{\infty} \frac{1 - \frac{z}{\rho_n}}{1 - \frac{z}{\zeta_n}} \cdot \prod_{n=1}^{\infty} \frac{1 + \frac{z}{\hat{\rho}_n}}{1 + \frac{z}{\hat{\zeta}_n}}$$

where now $... < -\hat{\rho}_2 < -\hat{\zeta}_2 < -\hat{\rho}_1 < -\hat{\zeta}_1 < 0 < \zeta_1 < \rho_1 < \zeta_2 < \rho_2 < ...$, with $\rho_n = M + n - 1$ and $\hat{\rho}_n = G + (n-1)$ and the ζ_n's are the **zeros** of the meromorphic extension of $q - \psi(z)$ (equivalently the **poles** of $\frac{q}{q-\psi(z)}$) where $\psi(z) = \int_{-\infty}^{\infty}(e^{zx} - 1)\nu(x)dx$ in q7.

It turns out $\psi(z)$ can be computed explicitly as

$$\psi(z) = \frac{1}{\beta}(B(\alpha - \frac{z}{\beta}, 1-\lambda) - B(\alpha, 1-\lambda)) \; + \; \frac{1}{\beta_2}(B(\alpha_2 + \frac{z}{\beta_2}, 1-\lambda_2) - B(\alpha_2, 1-\lambda_2)) \tag{5}$$

where $B(x,y) = \Gamma(x)\Gamma(y)/\Gamma(x+y)$ and $\Lambda = 1+Y$, $\beta = 1$, $\alpha = M$, $\lambda_2 = 1+Y$, $\beta_2 = 1$, $\alpha_2 = G$. For each $q > 0$, we then solve for these roots numerically, which then gives us the two WH factors. This avoids the use of the double integral formula in Hwk 2 q1 to compute $\mathbb{E}(e^{iz\bar{X}_{e_q}})$.

Remark 0.1 (see Mathematica sheet at `https://martinforde.github.io/KyprianouMeromorphicNumericsWorks.nb`)

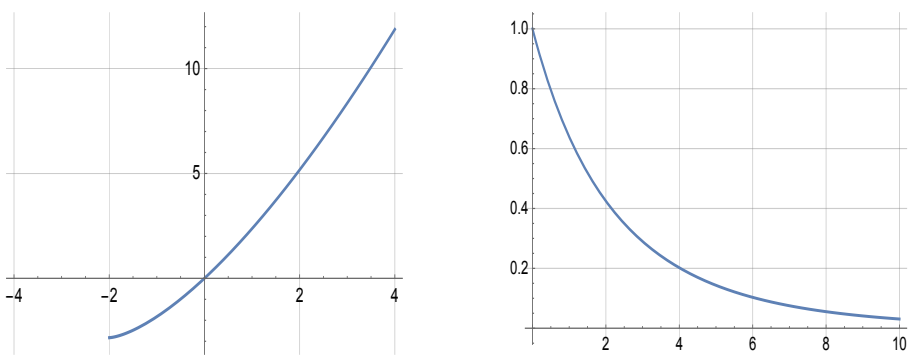

Figure 1: $V(p)$ and its inverse $\Phi(q)$ for q6 with $Y = 3/2$ and $G = 2$

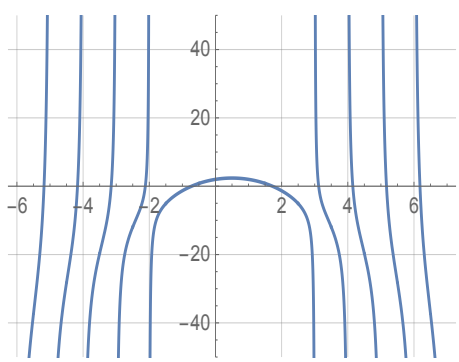

Figure 2: Here we see the poles and roots of $q - \psi(z)$ for q8 for $Y = 3/2$, $M = 3$, $G = 2$ and $q = 2$ using (5).

9. Compute the mgf of an Exp(q) random variable e_q. What can we say about the answer for $z \geq q$?

Solution. The mgf is

$$\mathbb{E}(e^{ze_q}) \quad = \quad f(z)$$

for $z < q$, and infinity otherwise, where $f(z) = \frac{q}{q-z}$. Clearly the function f has pole at $z = q$. Nevertheless, we can still consider the function f for all $z \in \mathbb{C}$ (this is known as the *meromorphic extension* of $\frac{q}{q-z}$ defined on $\{z < q\}$, even though it is not the mgf of e_q in this region.